<u>BCSE308P - Computer Networks Lab</u>
<u>ASSESSMENT – 5(L9+L10)</u>
<u>Socket Programming - TCP & UDP</u>

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Socket programming enables processes to communicate across a network.
It uses the **Transport Layer** protocols of the **TCP/IP model** to establish communication between client and server systems.

There are two major types of socket communication:

1. **TCP (Transmission Control Protocol):**
    - Connection-oriented protocol.
    - Ensures reliable, ordered, and error-checked delivery of data.
    - Requires a connection setup (handshake) before data transfer.
    - Example: Web browsing (HTTP).
2. **UDP (User Datagram Protocol):**
    - Connectionless protocol.
    - Faster but does not guarantee delivery or order.
    - Suitable for real-time applications like video streaming or gaming.

Difference between TCP and UDP

Feature	TCP	UDP
Type	Connection-oriented	Connectionless
Reliability	Reliable (acknowledgments, retransmission)	Unreliable
Speed	Slower (more overhead)	Faster
Ordering	Maintains packet order	No guaranteed order
Use Cases	Web, Email, File Transfer	Streaming, DNS, VoIP

TCP Client-Server Communication

TCP Server (tcp_server.c)

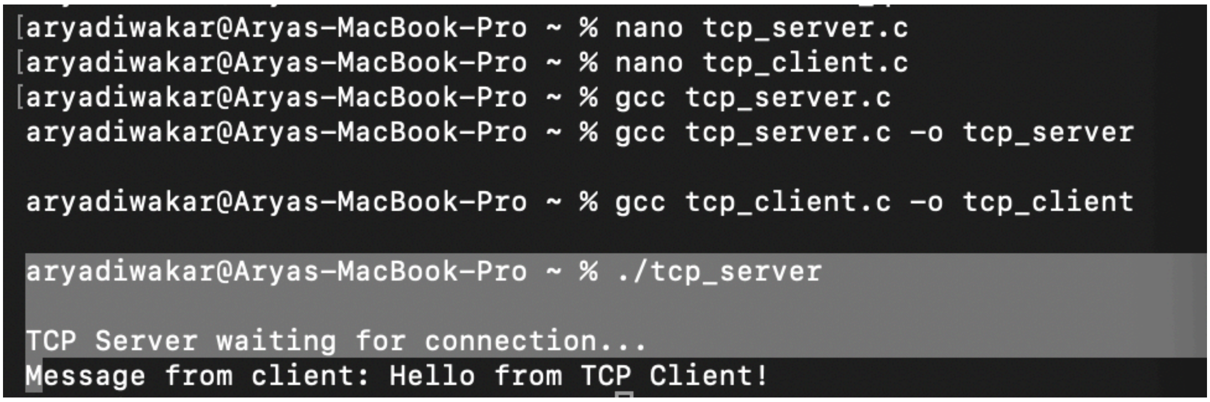
```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <arpa/inet.h>

int main() {
    int sockfd, newSock;
    struct sockaddr_in serverAddr, clientAddr;
    char buffer[1024];
    socklen_t addr_size;

    sockfd = socket(AF_INET, SOCK_STREAM, 0);
    serverAddr.sin_family = AF_INET;
    serverAddr.sin_port = htons(8080);
    serverAddr.sin_addr.s_addr = INADDR_ANY;

    bind(sockfd, (struct sockaddr*)&serverAddr, sizeof(serverAddr));
    listen(sockfd, 5);
    printf("TCP Server waiting for connection...\n");

    addr_size = sizeof(clientAddr);
    newSock = accept(sockfd, (struct sockaddr*)&clientAddr, &addr_size);
    recv(newSock, buffer, sizeof(buffer), 0);
    printf("Message from client: %s\n", buffer);

    strcpy(buffer, "Hello from TCP Server!");
    send(newSock, buffer, strlen(buffer) + 1, 0);

    close(newSock);
    close(sockfd);
    return 0;
}
```

TCP Client (tcp_client.c)

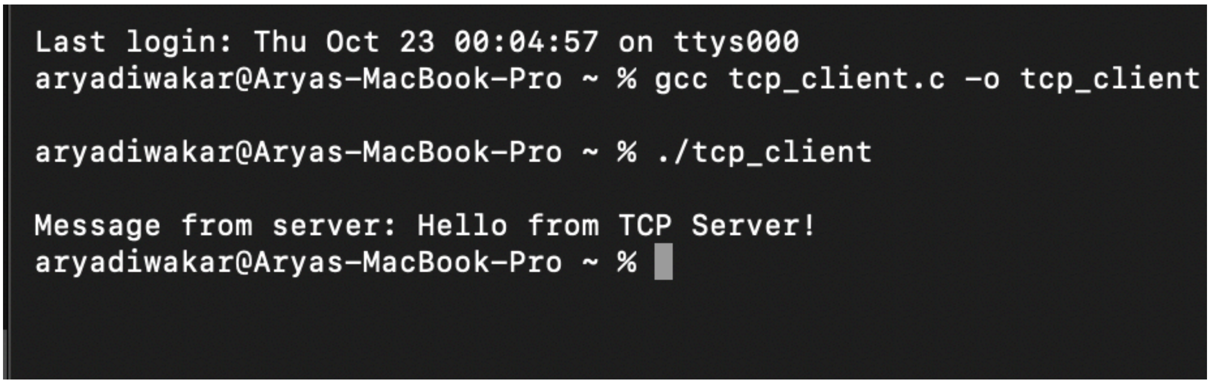
```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <arpa/inet.h>

int main() {
```

```
    int clientSocket;
    struct sockaddr_in serverAddr;
    char buffer[1024];

    clientSocket = socket(AF_INET, SOCK_STREAM, 0);
    serverAddr.sin_family = AF_INET;
    serverAddr.sin_port = htons(8080);
    serverAddr.sin_addr.s_addr = inet_addr("127.0.0.1");

    connect(clientSocket, (struct sockaddr*)&serverAddr, sizeof(serverAddr));
    strcpy(buffer, "Hello from TCP Client!");
    send(clientSocket, buffer, strlen(buffer) + 1, 0);

    recv(clientSocket, buffer, sizeof(buffer), 0);
    printf("Message from server: %s\n", buffer);

    close(clientSocket);
    return 0;
}
```

Server:



Client:



UDP Client-Server Communication

UDP Server (udp_server.c)

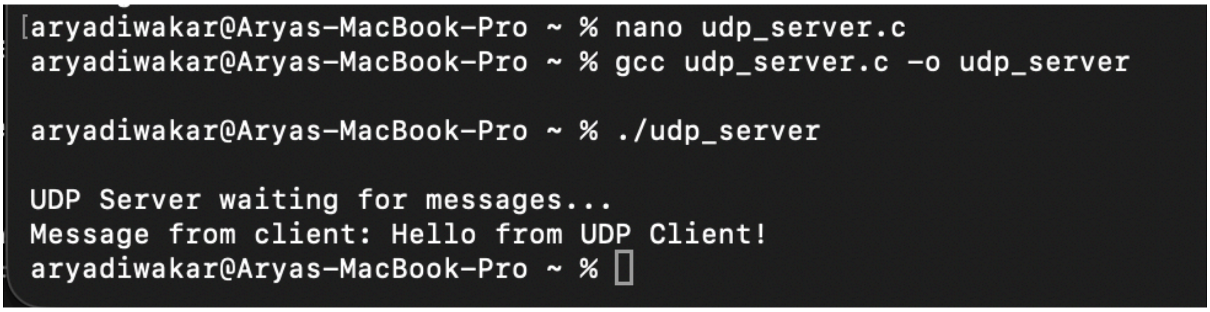
```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <arpa/inet.h>

int main() {
    int sockfd;
    struct sockaddr_in serverAddr, clientAddr;
    char buffer[1024];
    socklen_t addr_size;

    sockfd = socket(AF_INET, SOCK_DGRAM, 0);
    serverAddr.sin_family = AF_INET;
    serverAddr.sin_port = htons(8080);
    serverAddr.sin_addr.s_addr = INADDR_ANY;

    bind(sockfd, (struct sockaddr*)&serverAddr, sizeof(serverAddr));
    printf("UDP Server waiting for messages...\n");

    addr_size = sizeof(clientAddr);
    recvfrom(sockfd, buffer, sizeof(buffer), 0, (struct sockaddr*)&clientAddr, &addr_size);
    printf("Message from client: %s\n", buffer);

    strcpy(buffer, "Hello from UDP Server!");
    sendto(sockfd, buffer, strlen(buffer) + 1, 0, (struct sockaddr*)&clientAddr, addr_size);

    close(sockfd);
    return 0;
}
```

UDP Client (udp_client.c)

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <arpa/inet.h>

int main() {
```

```
    int clientSocket;
    struct sockaddr_in serverAddr;
    char buffer[1024];
    socklen_t addr_size;

    clientSocket = socket(AF_INET, SOCK_DGRAM, 0);
    serverAddr.sin_family = AF_INET;
    serverAddr.sin_port = htons(8080);
    serverAddr.sin_addr.s_addr = inet_addr("127.0.0.1");

    strcpy(buffer, "Hello from UDP Client!");
    addr_size = sizeof(serverAddr);
    sendto(clientSocket, buffer, strlen(buffer) + 1, 0, (struct sockaddr*)&serverAddr,
addr_size);

    recvfrom(clientSocket, buffer, sizeof(buffer), 0, (struct sockaddr*)&serverAddr,
&addr_size);
    printf("Message from server: %s\n", buffer);

    close(clientSocket);
    return 0;
}
```

Server:



Client: